3. The group algebra $A = k[G]$, where $G$ is a group. A representation of $A$ is the same thing as a representation of $G$, i.e., a vector space $V$ together with a group homomorphism $\rho : G \to \mathrm{Aut}(V)$, where $\mathrm{Aut}(V) = GL(V)$ denotes the group of invertible linear maps from the space $V$ to itself (the **general linear group** of $V$).

**Problem 2.3.15.** Let $V$ be a nonzero finite dimensional representation of an algebra $A$. Show that it has an irreducible subrepresentation. Then show by example that this does not always hold for infinite dimensional representations.

**Problem 2.3.16.** Let $A$ be an algebra over a field $k$. The center $Z(A)$ of $A$ is the set of all elements $z \in A$ which commute with all elements of $A$. For example, if $A$ is commutative, then $Z(A) = A$.

(a) Show that if $V$ is an irreducible finite dimensional representation of $A$, then any element $z \in Z(A)$ acts in $V$ by multiplication by some scalar $\chi_V(z)$. Show that $\chi_V : Z(A) \to k$ is a homomorphism. It is called the **central character** of $V$.

(b) Show that if $V$ is an indecomposable finite dimensional representation of $A$, then for any $z \in Z(A)$, the operator $\rho(z)$ by which $z$ acts in $V$ has only one eigenvalue $\chi_V(z)$, equal to the scalar by which $z$ acts on some irreducible subrepresentation of $V$. Thus $\chi_V : Z(A) \to k$ is a homomorphism, which is again called the central character of $V$.

(c) Does $\rho(z)$ in (b) have to be a scalar operator?

**Problem 2.3.17.** Let $A$ be an associative algebra, and let $V$ be a representation of $A$. By $\mathrm{End}_A(V)$ one denotes the algebra of all homomorphisms of representations $V \to V$. Show that $\mathrm{End}_A(A) = A^{\mathrm{op}}$, the algebra $A$ with opposite multiplication.

**Problem 2.3.18.** Prove the following "infinite dimensional Schur lemma" (due to Dixmier): Let $A$ be an algebra over $\mathbb{C}$ and let $V$ be an irreducible representation of $A$ with at most countable basis. Then any homomorphism of representations $\phi : V \to V$ is a scalar operator.

Hint: By the usual Schur's lemma, the algebra $D := \mathrm{End}_A(V)$ is an algebra with division. Show that $D$ is at most countably dimensional. Suppose $\phi$ is not a scalar, and consider the subfield $\mathbb{C}(\phi) \subset D$.